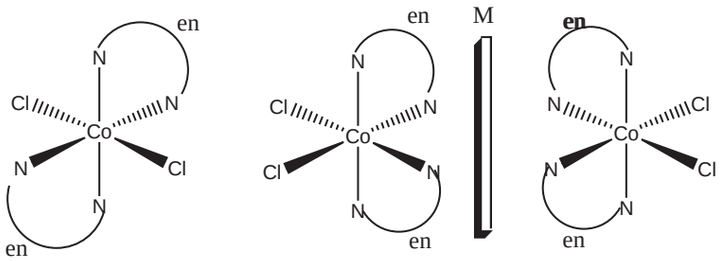
$[-1.2+1.2]\Delta_0 = 0$

9. Answer : option(a)

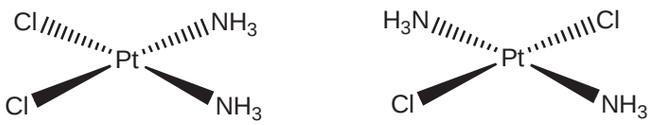
In all the complexes, the central metal ion is Co^{3+}, among the given ligands CN^- is the strongest ligand, which causes large crystal field splitting i.e maximum $\varnothing_0$

10. Answer : option(b)



Complexes given in other options (a), (c) and (d) have symmetry elements and hence they are optically inactive.

11. Answer : option(d)



12. Three isomers. If we consider any one of the ligands as reference (say Py), the arrangement of other three ligands (NH_3, Br^- and Cl^-) with respect to (Py) gives three geometrical isomers.

13. Answer : option(c)

(a)coordination isomers

(b) no isomerism (different molecular formula)

(c) $\leftarrow NCS$, $\leftarrow SCN$ coordinating atom differs : linkage isomers

14. Answer : option(a)

For $[MA_4B_2]^{n+}$ complexes geometrical isomers possible

$\left.\begin{matrix}[Co(NH_3)_4Br_2]Cl \\ [Co(NH_3)_4 Br\ Cl]Br\end{matrix}\right\} \rightarrow$ ionisation isomers

15. Answer : option(d)

Option (a) and (b) – geometrical isomerism is possible

Option (c) – ionization isomerism is possible

Option (d) – no possibility to show either constitutional isomerism or stereo isomerism

16. Answer : option(c)

(a) Fe^{2+} (b) Fe^{3+} (c) Fe^0

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